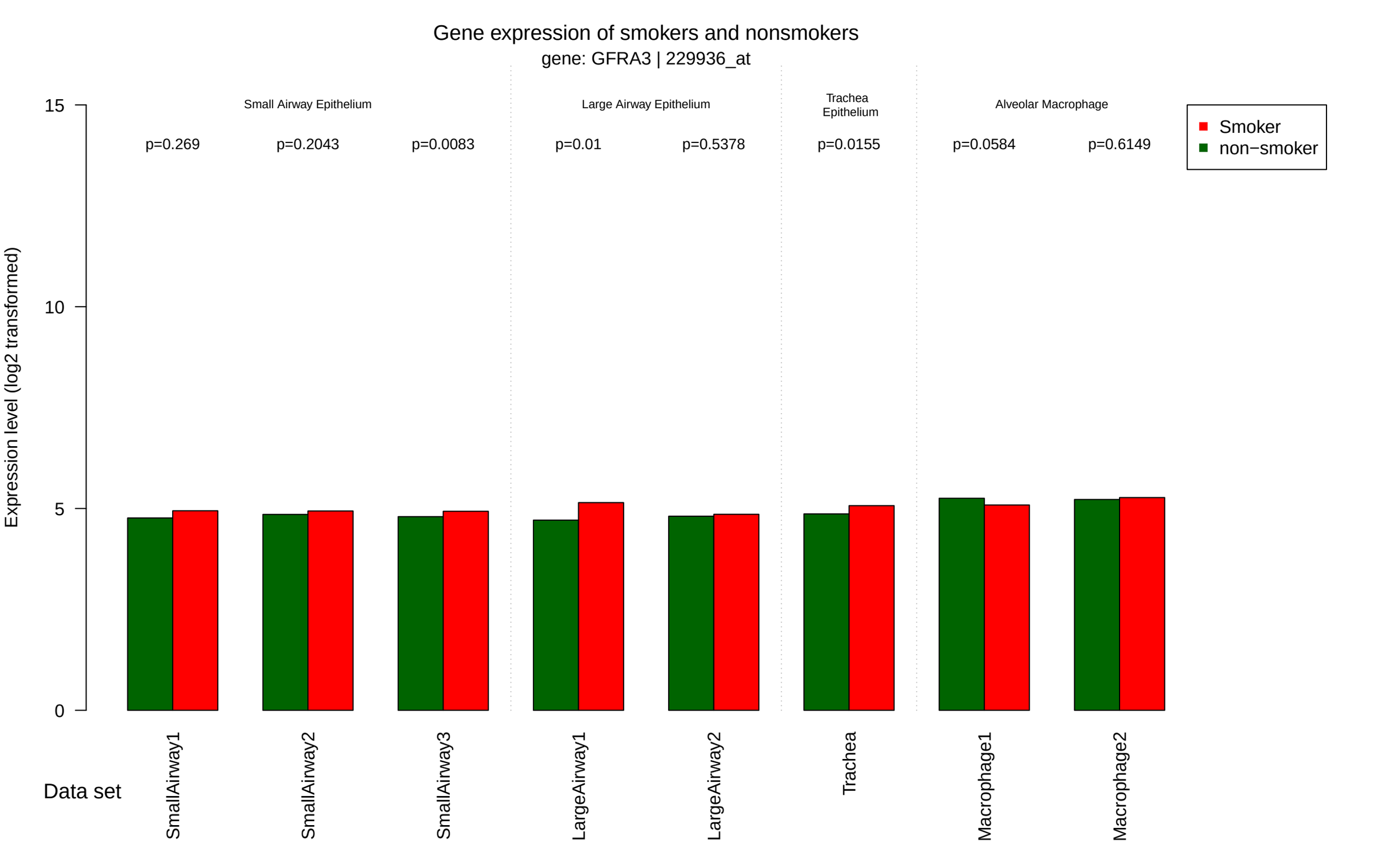
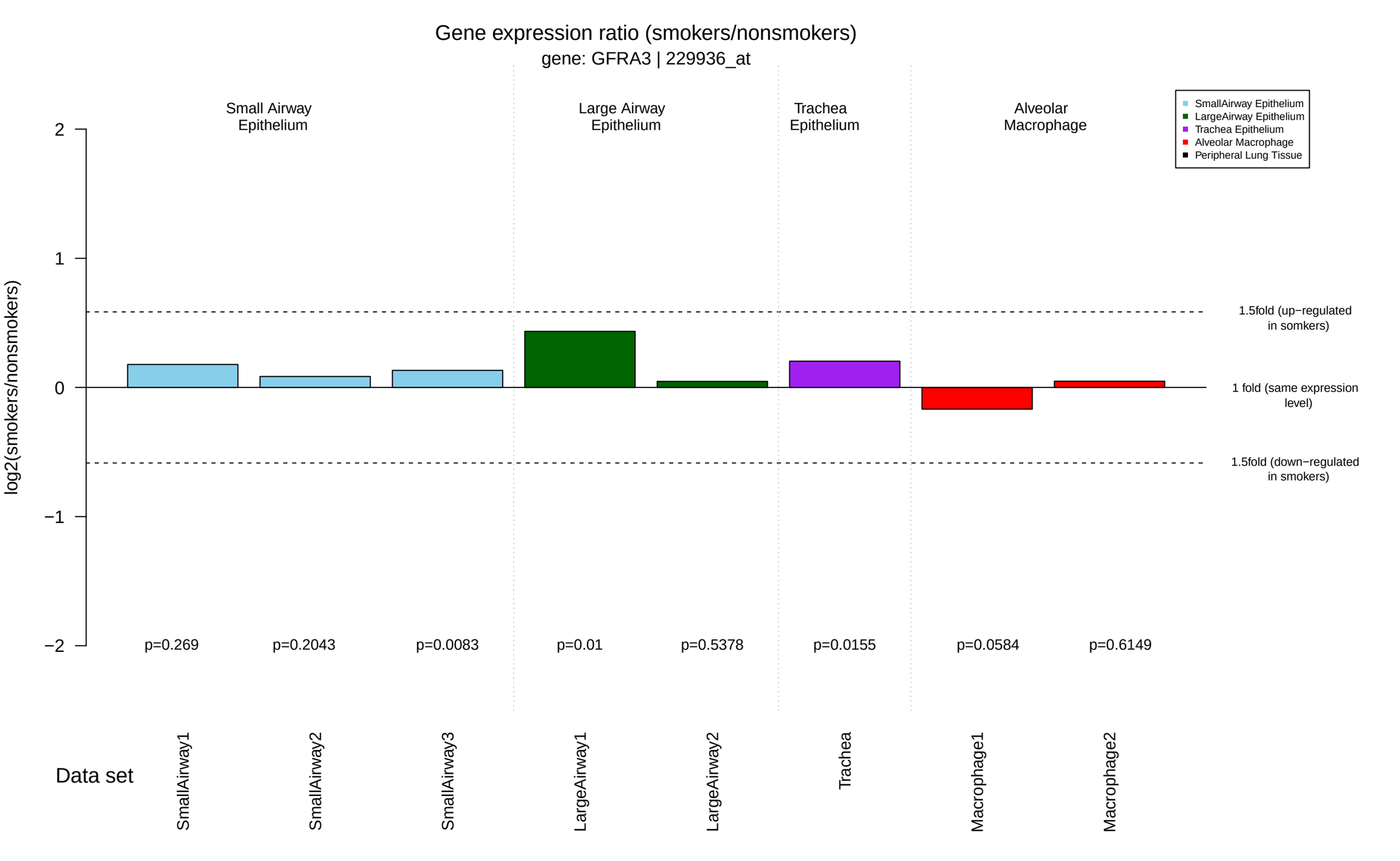
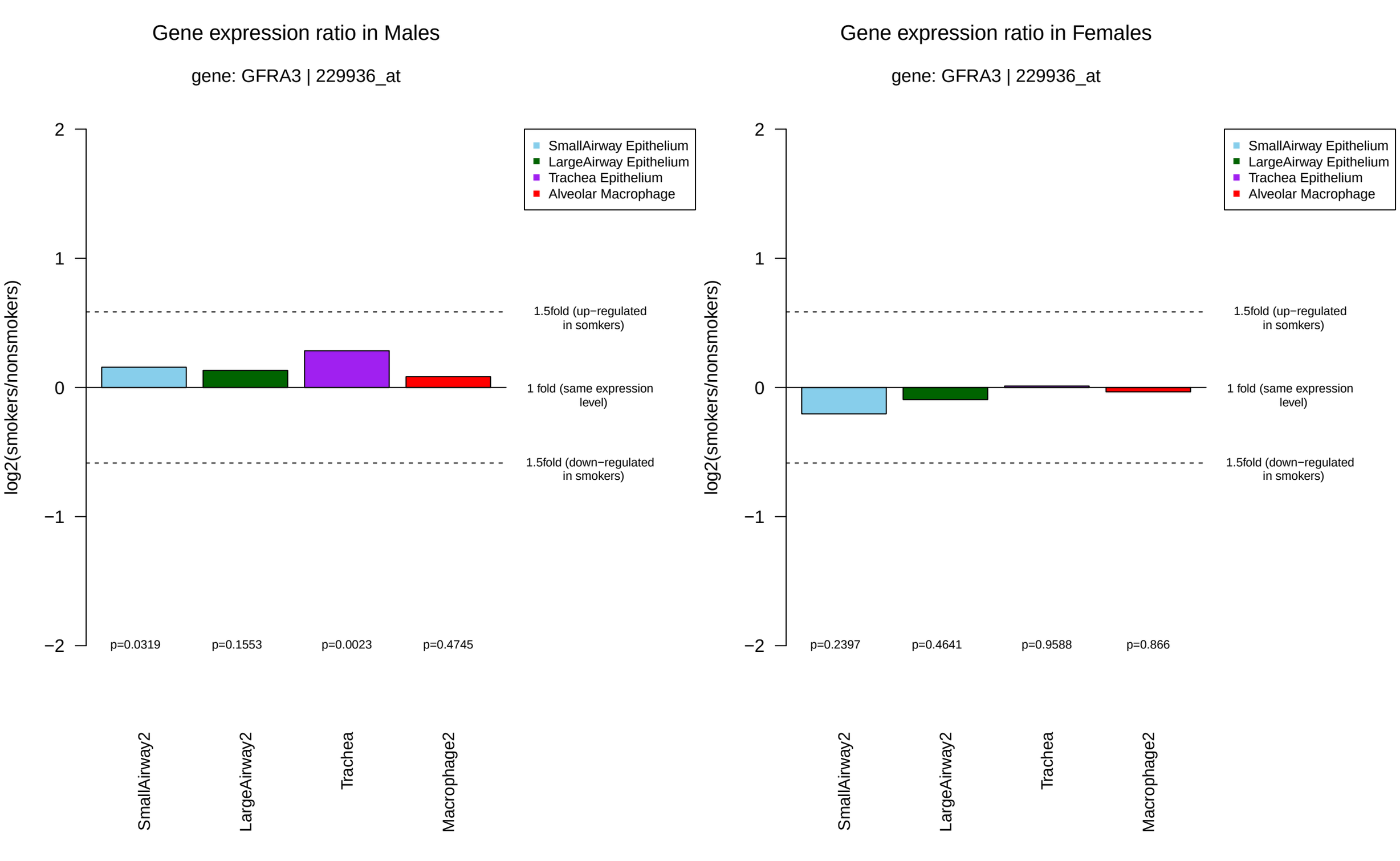
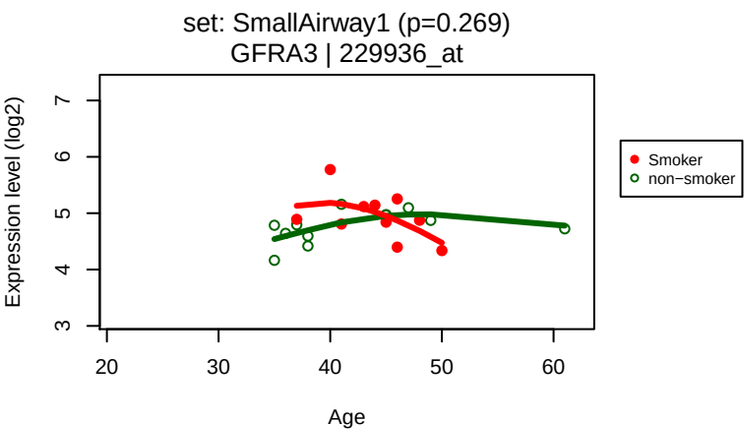
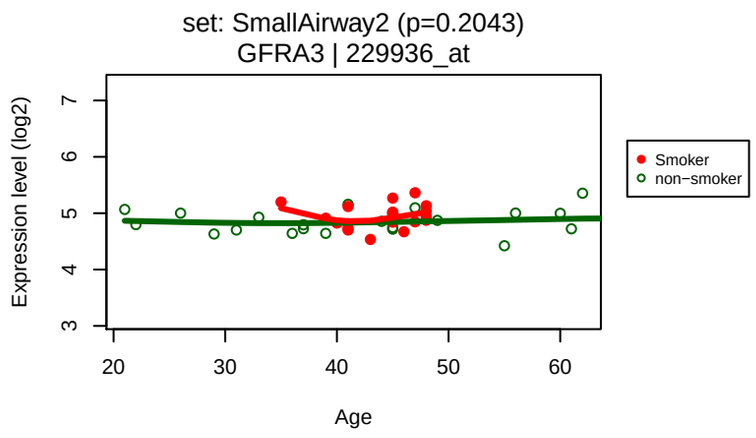
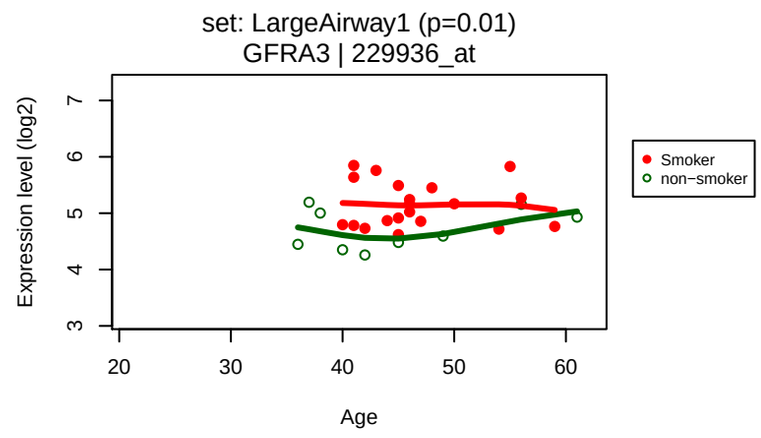

Gene expression of smokers and nonsmokers
gene: GFRA3 | 229936_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.269
p=0.2043
p=0.0083
p=0.01
p=0.5378
p=0.0155
p=0.0584
p=0.6149
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

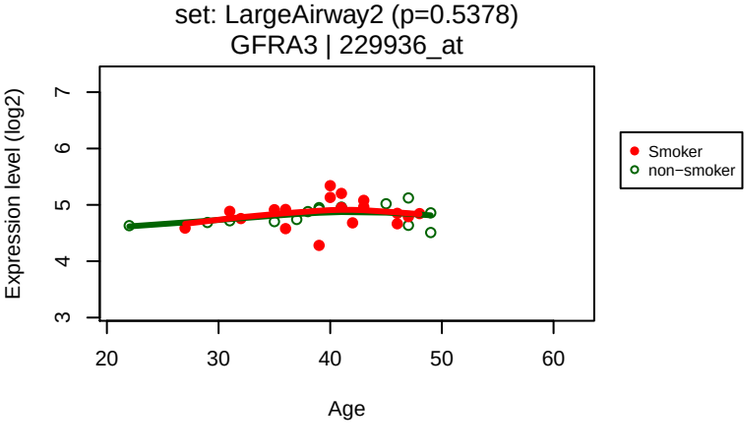

Gene expression ratio (smokers/nonsmokers)
gene: GFRA3 | 229936_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.269
p=0.2043
p=0.0083
p=0.01
p=0.5378
p=0.0155
p=0.0584
p=0.6149
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

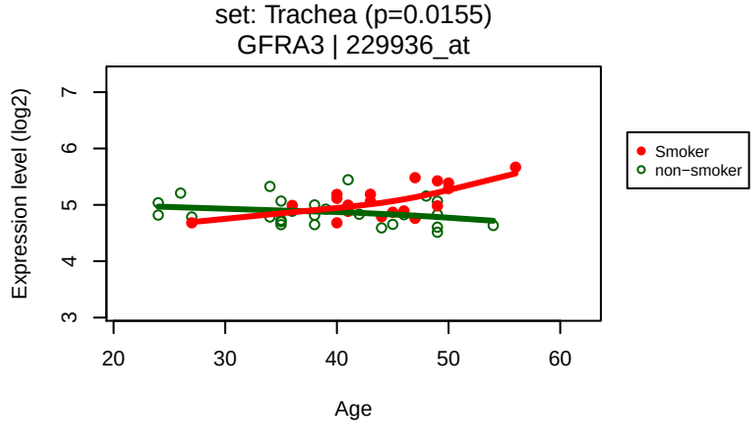

Gene expression ratio in Males
gene: GFRA3 | 229936_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0319
p=0.1553
p=0.0023
p=0.4745
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GFRA3 | 229936_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2397
p=0.4641
p=0.9588
p=0.866
SmallAirway2
LargeAirway2
Trachea
Macrophage2

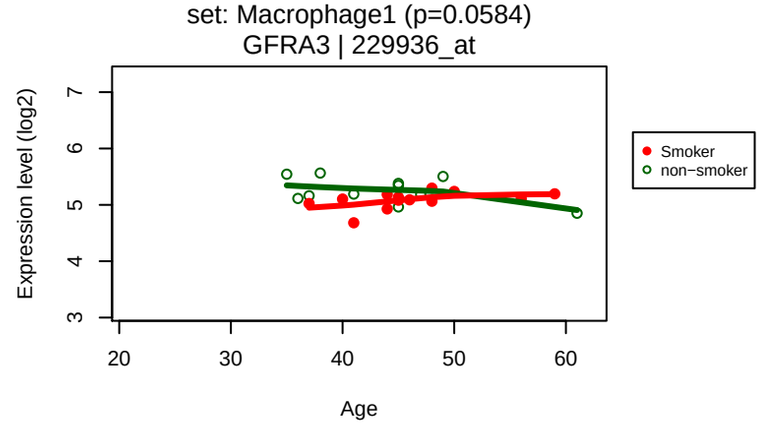

set: SmallAirway1 (p=0.269)
GFRA3 | 229936_at
Expression level (log2)
Age
Smoker
non−smoker

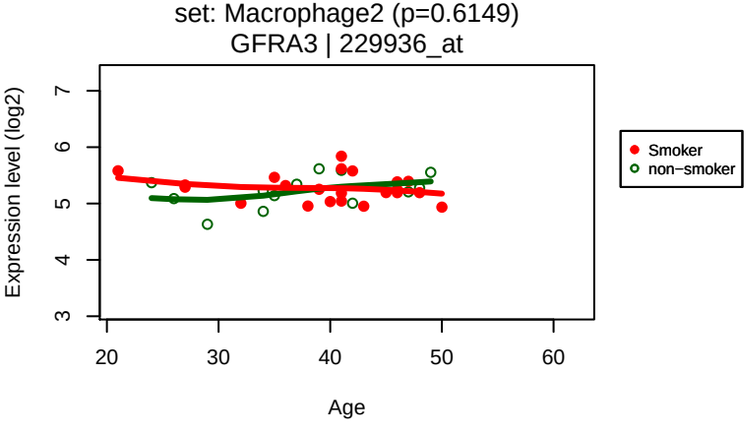

set: SmallAirway2 (p=0.2043)
GFRA3 | 229936_at
Expression level (log2)
Age
Smoker
non−smoker

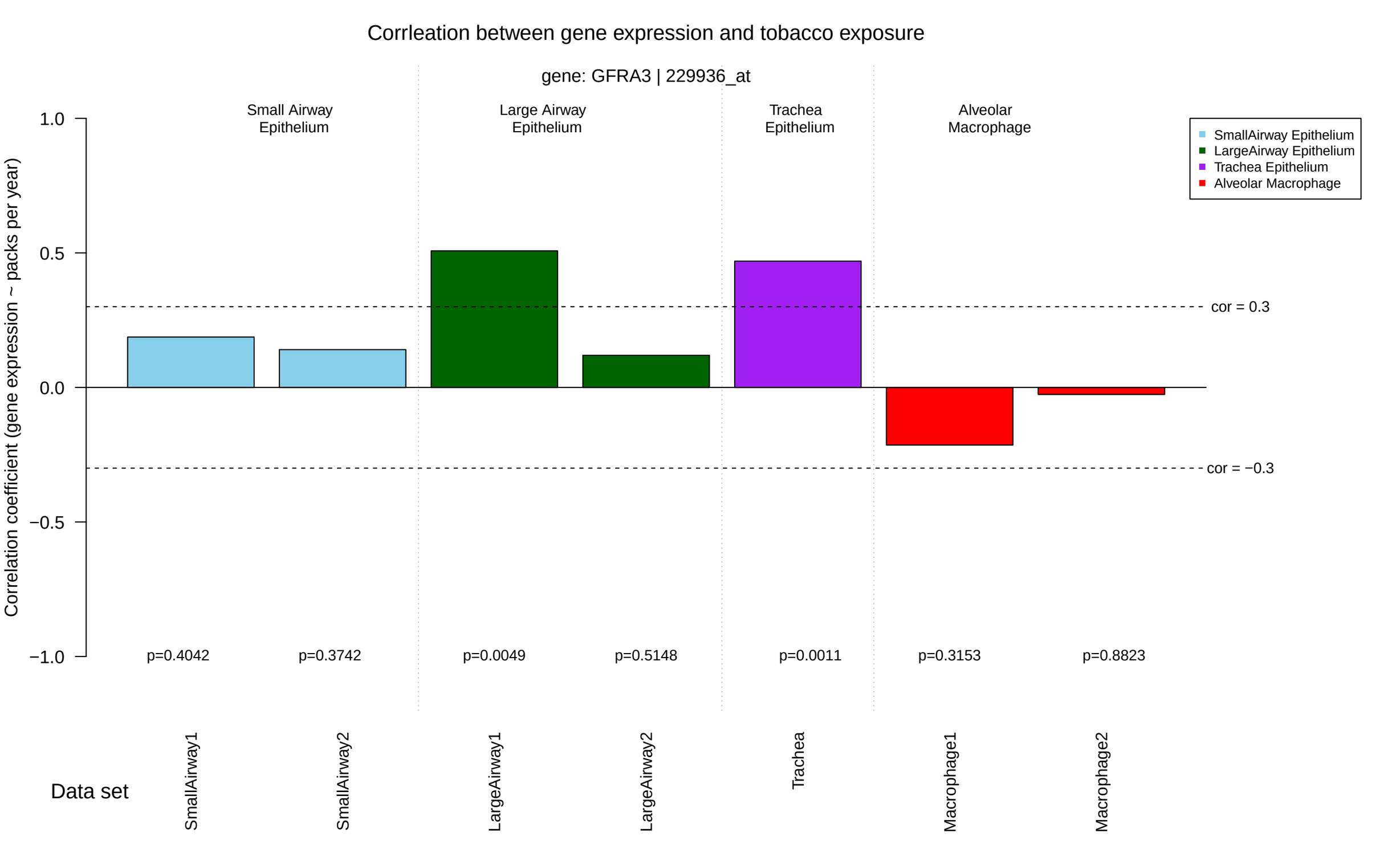

set: LargeAirway1 (p=0.01)
GFRA3 | 229936_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5378)
GFRA3 | 229936_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0155)
GFRA3 | 229936_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0584)
GFRA3 | 229936_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6149)
GFRA3 | 229936_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GFRA3 | 229936_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4042
p=0.3742
p=0.0049
p=0.5148
p=0.0011
p=0.3153
p=0.8823
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2